Lecture V
Spatial Linear Models

STAT 679: Statistical Methods for Spatial Lattice Data-Spring 2026
University of Wisconsin-Madison

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Outline

1. Guidelines for Linear and Spatial Regression
2. R-Demo
3. Assignment 5

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(1) General Guidelines for Traditional Linear Regression

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Key Takeaway from EDA

EDA helps us understand the structure of the data before formal modeling.

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3. **Assess whether variable transformations or variance weighting are needed**.
4. **Identify influential observations and/or outliers, then test and address them.**

(2) General Guidelines to Spatial Regression

(2a) After fitting and checking the full model

We can gradually reduce the model to the most essential components.

- Use expert knowledge of the topic.
- Use insights from exploratory data analysis.
- Use formal statistical inference, such as:
  - hypothesis testing,
  - confidence intervals,
  - coefficient significance,
  - model comparison.

Main idea

The final model should be statistically sound, interpretable, and supported by subject-matter knowledge.

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- In lattice data analysis, neighborhood definition is a key modeling decision.
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Why simpler is often better

Including more neighbors can make the analysis more complicated and increase computation time.

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Use expert and/or scientific knowledge

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Example: Bird species on islands

Suppose we study bird species on 10 islands and know that a specific species never leaves the island where it was born.

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- Then we can study the robustness of our conclusions.

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